

Helmholtz-coil alignment: beam bending

Bending of the electron beam in the Møller polarimeter because of Helmholtz coils **rotation** .
Møller Quads are **Turned ON** : Q1=1.46 Q2=0 Q3=0.29 Dipole=1.49

Coils	Beam	Beam shift at the distance from target			Beam shift at the distance from target		
Rotation	Shift	5.5m	9.0m	14.0m	5.5m	9.0m	14.0m
around	direction	cm/mrad	cm/mrad	cm/mrad	cm/m*mrad	cm/m*mrad	cm/m*mrad
X	X	+0.09	+0.14	+0.22	+0.016	+0.016	+0.016
X	Y	+0.07	+0.11	+0.17	+0.013	+0.012	+0.012
Y	X	+0.03	+0.06	+0.09	+0.006	+0.007	+0.006
Y	Y	-0.23	-0.40	-0.65	-0.042	-0.044	-0.046

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Shift	Shift	5.5m	9.0m	14.0m	5.5m	9.0m	14.0m
along	direction	cm/mm	cm/mm	cm/mm	cm/m*mm	cm/m*mm	cm/m*mm
X	X	+0.04	+0.07	+0.12	+0.007	+0.008	+0.009
X	Y	+0.18	+0.31	+0.50	+0.033	+0.034	+0.038
Y	X	-0.11	-0.19	-0.32	-0.020	-0.021	-0.023
Y	Y	+0.10	+0.17	+0.26	+0.018	+0.019	+0.019
X+Y	X	-0.09	-0.17	-0.28	-0.016	-0.019	-0.020
X+Y	Y	+0.24	+0.42	+0.68	+0.044	+0.046	+0.049

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Rotation	Shift	5.5m	9.0m	14.0m	5.5m	9.0m	14.0m
around	direction	cm/mrad	cm/mrad	cm/mrad	cm/m*mrad	cm/m*mrad	cm/m*mrad
X	X	+0.14	+0.23	+0.35	+0.02	+0.026	+0.025
X	Y	+0.03	+0.04	+0.07	+0.055	+0.0044	+0.005
Y	X	+0.03	+0.04	+0.07	+0.0055	0.0044	0.005
Y	Y	-0.14	-0.23	-0.35	-0.02	-0.026	-0.025

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Shift	Shift	5.5m	9.0m	14.0m	5.5m	9.0m	14.0m
along	direction	cm/mm	cm/mm	cm/mm	cm/m*mm	cm/m*mm	cm/m*mm
X	X	+0.055	+0.08	+0.12	+0.010	+0.009	+0.009

X	Y	+0.125	+0.22	+0.35	+0.023	+0.024	+0.025
Y	X	-0.125	-0.22	-0.35	-0.023	-0.024	-0.025
Y	Y	+0.055	+0.08	+0.12	+0.010	+0.009	+0.009
X+Y	X	-0.09	-0.16	-0.23	-0.016	-0.017	-0.016
X+Y	Y	+0.17	+0.28	+0.44	+0.031	+0.031	+0.031
X+Y	X for 1Tl	-0.035	-0.055	-0.085	-0.006	-0.006	-0.006
X+Y	Y for 1Tl	+0.040	+0.060	+0.095	+0.007	+0.007	+0.007